

Workbook: The Mathematical Foundations of Basic Electronics

Objective: Before we study circuits, we must be familiar with the "language" of the graphs that describe them. Use [Desmos.com](https://www.desmos.com) to complete these investigations.

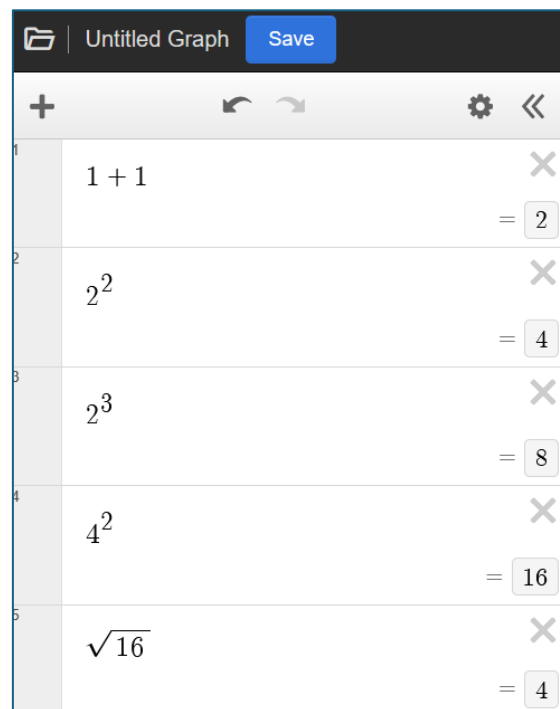
Don't worry about the technical terms, we are going to address them following this.

Practical Workbook Exercises: From Math to Electronics

1. Single Calculations (Calculator Check)

Get familiar with entering basic functions into the input line.

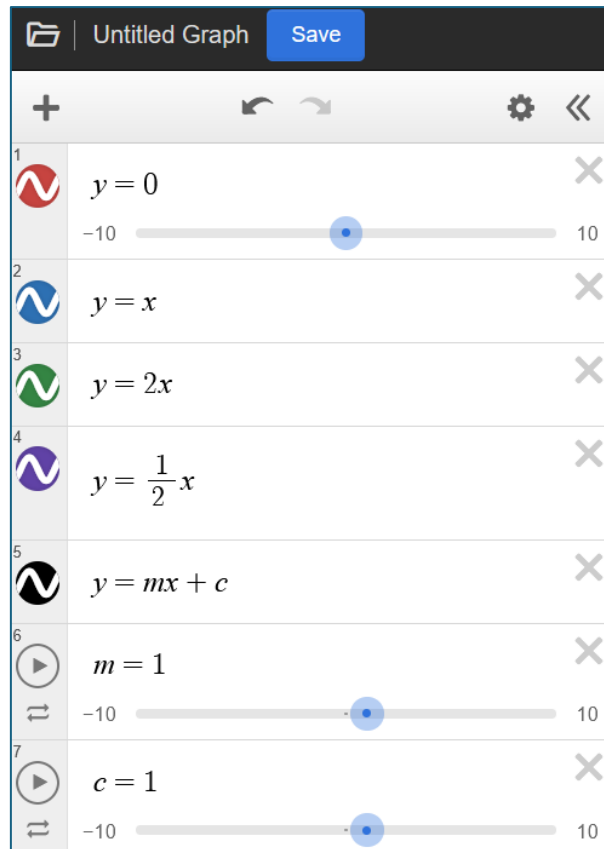
- $1 + 1$
- 2^2
- 2^3
- 4^2
- $\sqrt{16}$ (Note: Desmos automatically converts $\sqrt{16}$ into a square root symbol)



2. Linear World (Straight Lines & Ohm's Law Connection)

Explore how slope and intercept change the line.

- $y = 0$ (Note: Press Enter to add a sliders for y value constant)
- $y = x$
- $y = 2x$
- $y = \frac{1}{2}x$ (Note: Desmos automatically converts the fraction)
- $y = mx + c$ (Note: Click the prompt to add "all" sliders for m and c)



- **Question:** What happens to the steepness of the curve when you make m (the slope) larger versus smaller?
- **Question:** What happens to the y-intercept (where the line cuts the y-axis) of the curve when you make c (the offset) larger versus smaller?

3. Growth & Decay (Capacitor Charging)

Observe how a curve rises quickly and then levels off.

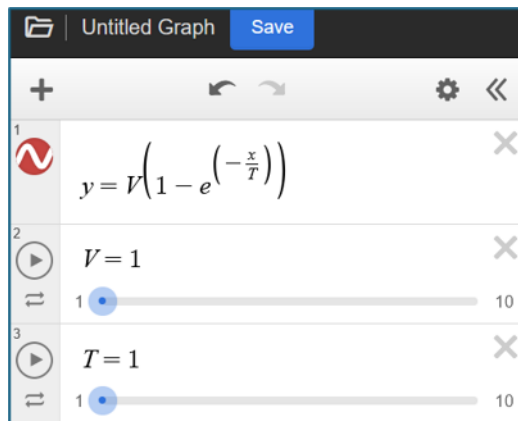
Setup

Click on the Spanner  (Settings) on the right of the screen and change the display x and y axis:

☒ X-Axis
 0 ≤ x ≤ 10

☒ Y-Axis
 -8 ≤ y ≤ 10

- $y = V * (1 - e^{(-x / T)})$ (Note: Click to add sliders for V and T) (Set both V and T from 1 to 10)

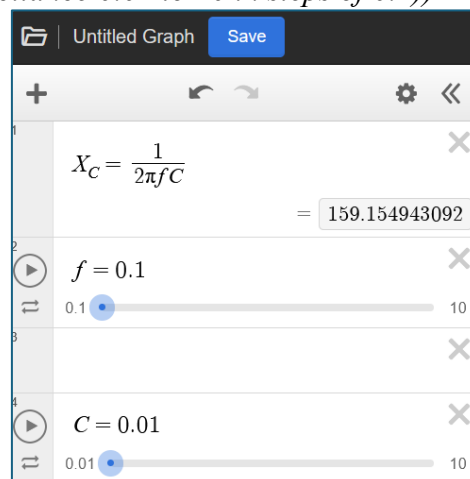


- **Question:** What happens to the height of the curve when you make V (the amplitude) larger versus smaller?
- **Question:** What happens to the steepness of the curve when you make T (the time constant) larger versus smaller?

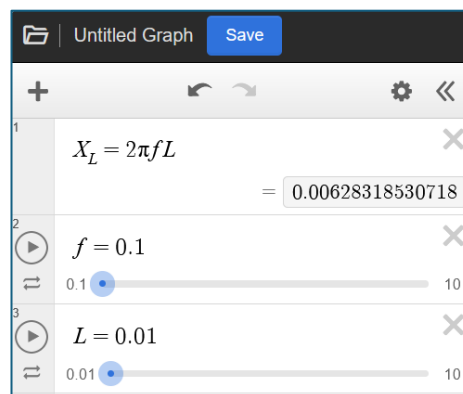
4. Impedance & Frequency (Reactance Formulas)

Explore how frequency affects components.

- **Capacitive Reactance:** $X_C = 1 / (2 * \pi * f * C)$ (Note: Add sliders for f (frequency 0.1 to 10) and C (capacitance 0.01 to 10 in steps of 0.1))



- **Inductive Reactance:** $X_L = 2 * \pi * f * L$ (Note: Add sliders for f and L (inductance))



- **Question:** Notice what happens to X_C versus X_L as you increase frequency (f). Which one drops to zero, and which one shoots upward?

5. AC Waves (Sinusoidal Functions)

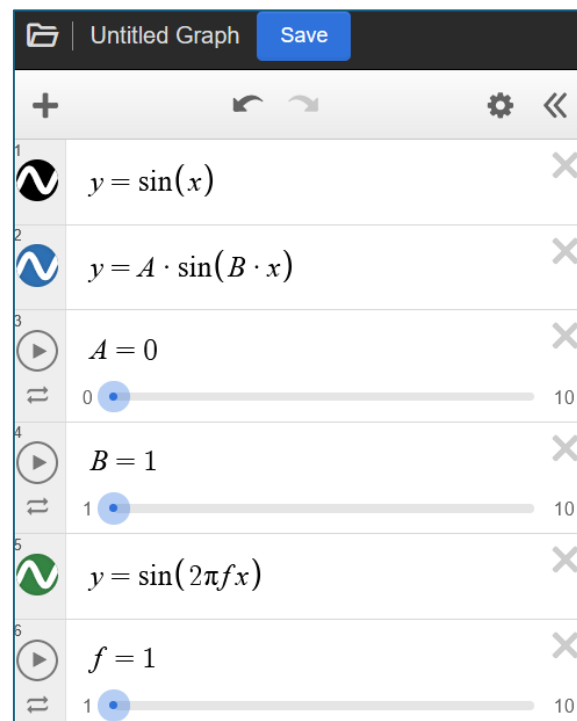
Visualize alternating current and its properties.

Setup

Click on the Home (house icon) on the right of the screen to centre the x and y axis:



- $y = \sin(x)$
- $y = A \cdot \sin(B \cdot x)$ (Note: Add sliders for A (amplitude/peak voltage 0 to 8) and B (frequency factor 1 to 10))
- $y = \sin(2 \cdot \pi \cdot f \cdot x)$ (Note: Add slider for f to see actual frequency cycles 1 to 10)



- **Question:** What happens to the height of the curve when you make A (amplitude) larger versus smaller?
- **Question:** What happens to the number of waves of the curve when you make B or f (frequency) larger versus smaller?